

Maqueta: Height Provenance in Open-Data 3D Urban Reconstruction Across 118 Places

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CAOS open-research programme · CAOS_RES_Maqueta, open-data 3D reconstruction of real places

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_RES_Maqueta

Abstract. A 3D city model is only as accurate as its building heights, and a model built from open data mostly infers those heights, from a mapped floor count, a height raster or a default value, rather than taking them from a mapped height. Maqueta reconstructs real places from open geospatial layers, with terrain from the Copernicus GLO-30 surface model, building footprints and roads from Overture Maps, water, green areas and rail from OpenStreetMap, population from GHS-POP, and building heights from the Google Open Buildings 2.5D raster where it has coverage, and it records for every building how its height was obtained. Across the 118 baked places and 4.05 million buildings, 8.8% of heights come from a mapped height attribute, 6.4% from a mapped floor count, 73.2% from the Open Buildings 2.5D height raster, itself a model estimate from satellite imagery, and 11.6% from a default prior of 8 m. A public LoD2 reference exists for two of the places, both in the Netherlands, where the raster has no coverage. Against the 3D BAG, the fused heights have a root-mean-square error of 3.1 m at Delft, with a bias of -0.1 m, and of 8.3 m at Amsterdam, with a bias of -6.6 m; the Amsterdam figure rests on 50 reference buildings. The raster heights, which supply most of the buildings, are not validated by this benchmark, and public ground truth for open-data building heights is scarce.

Index Terms. 3D city models, geospatial data fusion, building height, LoD2, open data, data provenance, OpenStreetMap, Google Open Buildings.

I. INTRODUCTION

A 3D city model is used for solar-potential studies, flood and wind simulation, visibility and noise analysis, and urban planning [1], and every one of those uses depends on the building heights. Authoritative models at level of detail 2 (LoD2), reconstructed from airborne lidar, exist for a handful of well-mapped countries [3]. For most of the world the heights are not surveyed: they are estimated from attributes such as a floor count, from a height raster, or from a default value, and generating 3D city models without elevation data is a matter of such estimation [2]. When a single-source extrusion is rendered as a 3D city, nothing in the model distinguishes an estimated height from a mapped one.

Maqueta is a multi-source fusion tool that reconstructs an area from open layers, each carrying its source, licence, release

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Interactive tool and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_RES_Maqueta; the geoscena fusion core at https://github.com/fsantibanezleal/CAOS_GeoScena.

TABLE I
LAYERS OF THE 118 BAKED PLACES: THE NUMBER OF PLACES THAT CARRY EACH LAYER.

Layer	Source	Places
Terrain	Copernicus GLO-30	118
Population	GHS-POP R2023A	116
Buildings, roads	Overture Maps	100
Green areas	OpenStreetMap	26
Water	OpenStreetMap	24
Rail	OpenStreetMap	23
LoD2 reference	3D BAG	2

and fetch date, and it records for every building how its height was obtained. The provenance is read here across the 118 baked places, and the heights are benchmarked where an authoritative LoD2 reference exists. The contribution is the per-building height provenance and its measurement across the places, not a new reconstruction algorithm.

II. SOURCES AND THE HEIGHT LADDER

Layers. Terrain comes from the Copernicus GLO-30 surface model [5], building footprints and roads from Overture Maps [7], which conflates OpenStreetMap, national mapping agencies and machine-learned roofprints, water, green areas and rail from OpenStreetMap [8], and population from the GHS-POP R2023A grid [6]. The layers present depend on the place (Table I): every place has terrain, 116 have population, 100 have buildings and roads, 23 to 26 have the OpenStreetMap context layers, and the 18 places without buildings, landscapes and heritage sites such as the Grand Canyon, Machu Picchu and the Chuquicamata mine, have terrain and, at 16 of them, population. The two Dutch places add the LoD2 reference of the 3D BAG [3] as a layer of its own. Every building also carries a land-cover class from ESA WorldCover 2021 [9], the 10 m class sampled at a point inside its footprint, which the tool uses to colour and aggregate buildings by land cover; unlike the layers, this attribute has no provenance entry in the place manifests.

Height ladder. Each building footprint takes its height from the first of four rungs that yields a positive value, and the rung is recorded with the building: (1) a mapped height attribute in Overture or OpenStreetMap, used directly; (2) a mapped `num_floors` attribute, multiplied by 3.2 m; (3) the Google

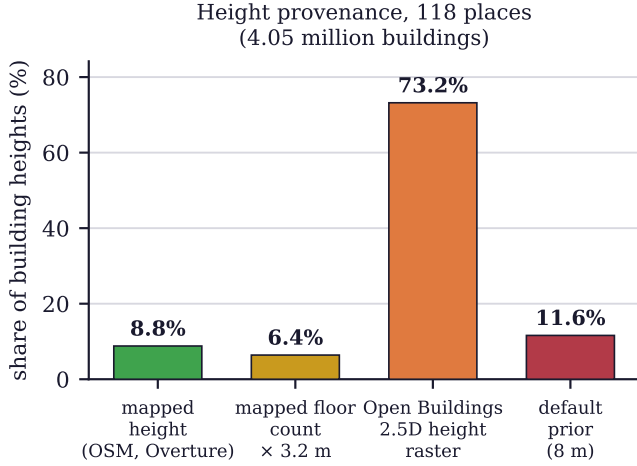


Fig. 1. Height provenance of the 4.05 million buildings of the 118 baked places: 8.8% from a mapped height attribute, 6.4% from a mapped floor count, 73.2% from the Open Buildings 2.5D height raster and 11.6% from the default prior.

Open Buildings 2.5D height raster, sampled at the footprint where the raster has coverage (Latin America, Africa, and South and Southeast Asia); and (4) a default prior of 8 m. Heights are clipped below at 2.5 m. The raster is a model estimate from Sentinel-2 imagery, for which its producers report a mean absolute error of 1.5 m [4]; only the first rung is a mapped value, and none is a survey measurement made for this work.

III. HEIGHT PROVENANCE ACROSS 118 PLACES

Over the 4.05 million buildings of the 118 places (Fig. 1), 8.8% of heights come from a mapped height attribute, 6.4% from a mapped floor count, 73.2% from the Open Buildings 2.5D raster and 11.6% from the default prior: more than nine in ten heights are estimated rather than mapped. The mix varies strongly between places, and the global shares depend on which places were baked. The raster is the dominant rung at 62 of the 100 places with buildings and is not used at 35 of them; a few large Latin American places carry most of its share, among them Buenos Aires (98.7% of 223 thousand buildings) and Santiago (93.9% of 179 thousand). Where the raster has no coverage, the prior can dominate, as in Tokyo (91.5%), and where mapped heights are abundant they dominate instead, as in New York (99.3%) and São Paulo (87.5%). At the median place with buildings, 0.3% of heights are mapped. Because the rung is recorded per building, a downstream user can filter or weight each height by how it was obtained.

IV. BENCHMARK WHERE A LOD2 REFERENCE EXISTS

The 3D BAG reconstructs LoD2 models of all buildings in the Netherlands from lidar [3], and it gives the reference height as the median roof height minus the ground level. Two of the 118 places, Delft and Amsterdam Centrum, lie in its coverage (Fig. 2). Each fused building is matched by its centroid to the nearest reference building within 25 m, so several fused footprints can share one reference building, and the error is

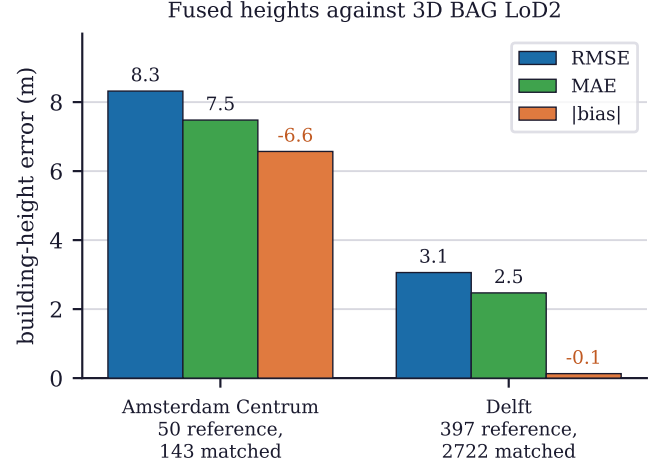


Fig. 2. Fused-height error against the 3D BAG LoD2 reference at the two places that have one. Each fused building is matched to the nearest reference building within 25 m. Root-mean-square error 3.1 m at Delft (bias -0.1 m; 2722 fused buildings matched to 397 reference buildings) and 8.3 m at Amsterdam (bias -6.6 m; 143 matched to 50).

taken over the matched fused buildings. At Delft, 2722 fused buildings (17.1% of the place) match 397 reference buildings, with a root-mean-square error of 3.1 m, a mean absolute error of 2.5 m and a bias of -0.1 m. At Amsterdam, 143 fused buildings (1.0% of the place) match 50 reference buildings, with 8.3 m, 7.5 m and a bias of -6.6 m. The heights of both places come almost entirely from floor counts and the default prior, since the raster does not cover Europe: 41% floor counts and 59% prior at Delft, 7% and 92% at Amsterdam. The larger error and the low bias at Amsterdam are consistent with the 8 m default lying below the typical height of the matched buildings, although two places, one of them with 50 reference buildings, cannot establish that relation. Neither place contains a raster height, so the rung that supplies 73.2% of all heights is not validated by this benchmark.

V. DISCUSSION AND SCOPE

An open-data 3D city model is mostly estimated: in these 118 places fewer than one building height in ten comes from a mapped value, and most come from a height raster that is itself a model output. Recording the rung per building makes that visible and lets a user treat a mapped height and a default differently. The validation is thin. Public LoD2 references cover two of the places, both in a country without raster coverage, so they test the floor-count and prior rungs only, and the Amsterdam estimate rests on 50 reference buildings. The global shares describe this set of places and would change with a different selection. The reconstructed scenes serve visualisation, teaching and planning context, not survey-grade modelling. The reusable fusion core, *geoscena*, turns an area of interest into a provenance-tracked scene bundle, and the product pipeline bakes a curated registry of places on top of it.

VI. CONCLUSION

Maqueta reconstructs 118 real places from open geospatial layers and records the provenance of every building height. Of 4.05 million heights, 8.8% come from a mapped height attribute, 6.4% from a mapped floor count, 73.2% from the Google Open Buildings 2.5D height raster and 11.6% from a default prior. Where a public LoD2 reference exists, at two Dutch places, the fused heights have a root-mean-square error of 3.1 and 8.3 m; those places test only the floor-count and prior rungs, and public ground truth for open-data building heights remains scarce.

APPENDIX A REPRODUCTION

The repository benchmark (`data/derived/benchmark.json`) carries, per place, the layer count, the triangle and size totals, the height-provenance counts per rung and, where a LoD2 reference exists, the matched counts, the root-mean-square error, the mean absolute error and the bias. Each place's manifest (`data/derived/<place>/manifest.json`) records every layer's source, URL, licence, release, fetch date and method; the per-building land-cover class of Section II is not recorded there. The figure script under `manuscripts/geo-reconstruction/figures/` reads `mq.json` in the data folder beside it, a copy of the benchmark figures. The `geoscena` fusion core and the product pipeline (`data-pipeline/`) rebuild a place from the recorded source releases; newer releases of the

open sources change the result. The scenes were baked with `geoscena` at commit `ab0bbf8` (14 July 2026), pinned in `data-pipeline/requirements-bake.txt`; its 0.1.0 release on PyPI lacks the population, Open Buildings, LoD2, Sentinel-2 and environment fetchers that the bake uses. The open layers are fetched under their own licences and are not redistributed.

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